Temple University
Japan Campus

"Train Travel" Feasibility Study

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Executive Summary

The '**Train Travel' Tracker App** is a digital companion designed to transform the traditional experience of Japanese train travel into a social, gamified journey. By merging the manual act of station stamp collecting with digital tracking, the project offers a modern "digital souvenir" for both dedicated train enthusiasts and general tourists.

Introduction

This project is a progressive web application (PWA) that addresses a gap in the current Japanese travel market. While navigation tools and complex RPG games exist, there is a lack of a simple, dedicated logbook for recording station visits, train sightings, and stamp collections. This investment opportunity leverages a cost-effective, scalable infrastructure to capture a growing market of rail enthusiasts and international travelers.

Company's Role and Strength

Our development team of Liam Persad (Team Leader and Front-end), Cassady Mead (Back-end and Database), and Soraha Ebara (Resource Gathering and Translation), possesses the necessary technical skills to execute the project. Our team has already established a proof-of-concept live site and a GitHub repository, and has demonstrated a steady development trajectory as per our Gantt Chart.

Need/Problem

Currently, travelers must choose between navigation-heavy apps that lack personal station logging history or complex RPG-style games, that may be too overwhelming for casual users. This includes physical stamp books that are prone to damage and difficult to search. There is a clear need for a searchable, digitally preserved record of travels that remains accessible across all devices.

Unique Solution

The app differentiates itself by focusing on a "middle ground", providing a social media-style travel logging experience without heavy game mechanics or mundane navigation interfaces.
Key value propositions include:

- **Line Completion Tracking**: Visual percentages showing progress on specific train lines.
- **Digital Stamp Collection**: A digital alternative to physical stamp books.
- **Centralized Log**: A single interface for station visits, train model sightings, and timestamps.

Proof

Market research indicates a stable environment for such services, with the Japanese railroad market projected to grow at a 6.7% CAGR through 2030. The project's feasibility is supported by this growing market.

Resources

The project utilizes a serverless architecture and open-source data to minimize overhead.

- **Data**: Static rail data sourced from WikiData, EkiData, and ODPT.
- **Software**: JavaScript, Hotwire Turbo, and Capacitor JS for mobile porting.
- **Infrastructure**: Google Firebase (Firestore and Authentication) and Google Maps Platform.

Competition

Unlike competitors like Google Maps, which lacks travel logs features, or Ekimemo!, which focuses on RPG elements, this app prioritizes casual travel logging and accessibility.

Marketing Strategy

The app will utilize SEO targeting (e.g., "Japan train app"), advertising on social media platforms, and organic social sharing of line completion maps and achievements to drive user growth.

Budget

Initial development relies on the Firebase Spark plan and Google Maps free tier. As the user base scales to 50,000 active users, monthly costs for mapping and backend services are projected between $1,700 and $3,500.

Timeline

The project is scheduled for completion in early April 2026.

- **Alpha Prototyping**: March 16 – March 25.
- **Beta Prototyping**: March 25 – April 1.
- **Final Testing**: April 1 – April 3.

Conclusion

The 'Train Travel' Tracker App is a unique project that digitizes and gamifies a popular Japanese hobby. The app has a well tested technical foundation, clear market demand, and a cost-effective scaling plan, making it feasible to become the primary digital train logbook for rail travel in Japan.

Description of Product/Service

This app is centered around its primary appeal of being a digital companion for Japanese train enthusiasts or tourists looking to record their train travel adventures. It is designed to transform and complement the manual act of train station stamp keeping into a social, gamified experience.

Main Features:

- **Gamification of Travel:** This app displays line completion percentages for each train line travelled, which mirrors the feeling of completing levels or missions in a video game.
- **Centralized Tracking:** This app consolidates three major aspects that matter to train travel hobbyists in a single interface, making it easier to manage to use and share with others. These are:
    1. Station Visit Logs
    2. Physical to Digital Stamp Collecting
    3. Social Travel Logging

Our marketability comes from targeting a middle ground in the current market, positioning our app between navigation-based tools, extreme enthusiast forums and fantasy-heavy games:

Category	Existing Solution	App Advantage
Navigation	Google Maps, Navitime, etc.	Features a persistent log and gamified history that navigation apps lack or do not focus on.
Gaming	Ekimemo!	Removes complex RPG mechanics and focuses on a simpler social media style travel logging experience
Physical	Paper Stamp Books	Offers a searchable and digitally preserved digital record of stamps that is easier to manage and share.

This app is able to target both the specialized 'train enthusiast' market as well as the thriving tourist market in Japan.

- The tourist market is specifically interested in following the large local 'hype' behind trains and train travel, apart from existing train enthusiasts. This is because as much of Japan relies on rail as their primary method of transportation, this app provides tourists with a 'digital souvenir' of their travels in Japan.
- The ability to access this app from any devices lowers the barrier to entry as users can easily access the app's features during their travels, on top of the social layer which is implemented which appeals to the users' familiarity with existing social media apps.

Technical Considerations

This app is designed as a progressive web application based on a cloud infrastructure, with available ports to Android and iOS using a conversion library. It is built using the following technologies:

Core Technology Stack

Programming Language	- JavaScript (with Swift for iOS / Kotlin for Android) and additional feature libraries.
Front-end Framework	- Hotwire Turbo for implementing a Single-Page Application Framework. - Capacitor JS for adapting the web app into native OS calls for mobile operating systems (Android / iOS).
Mapping Engine	- Google Maps JavaScript API used to render interactive maps, plot station markers and draw line overlays for train lines.
Back-end Infrastructure	- Google Firebase provides the foundation, which includes Firebase Authentication for user authentication and Cloud Firestore for storing global rail data and dynamic user records.

Materials and Digital Resources

Static Rail Data	A manual database will be compiled dynamically from the following sources: - WikiData: General information on stations, lines, station coordinates, etc. - EkiData – Specialized open database on Japanese Train Stations, Train Lines, Train Operators, etc. - ODPT (Public Transportation Open Data Center) – Global open database on generic transport information, including rail services in Japan.
Cloud Services	- Using Firebase for database hosting and management and user authentication.
API Access	- Google Maps JavaScript API provides the required map data for the app

Product Delivery and Distribution

Hosting	- The web app will be hosted via GitHub pages, accessible via a live URL
Distribution	- Users can access the app via any modern web browser, allowing flexibility between devices (laptop, phone, tablet, etc.)
Open-Source Access	- This app's source code is delivered and maintained through a GitHub repository.

Product/Service Marketplace

The project targets two primary segments that form our market.

- **Train Enthusiasts:** This group is highly interested in train spotting, Japanese train history, and travel logging. Their focus is around "collection" and "completion," such as visiting every station on a specific line or spotting rare train models.
- **Tourists:** This broader segment of the market includes travelers who use Japan's rail network as their primary mode of transport. The app serves as a "digital souvenir" or modern replacement/companion to traditional physical stamp books.

Market Depth and Condition

The market is currently split between two extremes, leaving a **"middle ground"** that this app intends to fill.

- **Middle Ground Target:** Currently, users must choose between purely navigation apps that lack personal history OR complex RPG games that may be too gamified for casual users, some of which are only in Japanese.
- **Transition to Digital:** This app aims to be a companion, if not a complete replacement for physical stamp books which are prone to damage and can be difficult to search for a specific stamp, where this app takes the opening for a searchable, shareable digital logbook.

Project Viability and Demand

The project's viability can be reiterated by highlighting the following:

- **Low Barrier to Entry:** The app is a web application, which makes it "easier for a larger market of users to pick up" compared to heavy mobile games which may have complex tutorials and storylines.
- **Gamification:** The "Line Completion" and "Stamp Collection" feature and the ability to challenge friends provide a social and game-like incentive that drives repeat usage and group interaction.
- **Cost-Effective Infrastructure:** Utilizing **Google Firebase's free-tier (Spark plan)** and the **Google Maps JavaScript API free-tier**, the project has no inherent costs but leaves room for it to be monetized if these tiers become insufficient.

Marketing Strategy

This app will be promoted using the following Digital & Direct Marketing Channels:

- **SEO (Search Engine Optimization):** Utilizing targeted keywords like "Japan rail completion," "station stamp log," and "train spotting app" to ensure the app appears high in online searches.
- **Online Communities:** Engaging in forums and social media groups dedicated to train spotting and Japanese tourism to reach the target demographic directly.
- **Social Media Sharing:** Using "Line Completion" map visualizations as shareable content to drive organic reach through "word-of-mouth" or online competition between friends.

SWOT Analysis

INTE	POSITIVE	NEGATIVE
	Strengths 1. *Hybrid approach combining navigation history, stamp collecting, and social media in one app.* 2. *Line completion and stamp collecting provide users a game-like experience.* 3. *Accessible from any device via web browser.* 4. *Uses Firebase Spark plan and Google Maps free tier, no initial cost.*	**Weaknesses** 1. *Static data must be manually collected from external sources.* 2. *Since the app is developed with HTML/JS and Firebase, which have limited scalability for real-time social interactions, the social media feature may become difficult to manage as the number of users grows.*
	Opportunities 1. *Japanese Location-Based Entertainment market valued at $310M, growing at 9.6% Compound Annual Growth Rate.* 2. *Japanese railroad market growing at 6.7% CAGR through 2030.* 3. *Large tourist market in Japan looking for digital travel souvenirs.* 4. *Physical stamp books can be easily damaged and are difficult to search.*	**Threats** 1. *Ekimemo! is an existing competitor in the gaming space.* 2. *Google Maps and Firebase costs can increase significantly as the number of users grows beyond the free tier.*

Organization/Staffing

The development team behind this app consists of three people, whose skills will be assigned as follows but not solely restricted to:

Person	Role
Liam Persad	Team Leader; Front-end Focus Developer
Cassady Mead	Back-end Focus Developer, Database Manager
Soraha Ebara	Resource Gathering, Japanese Translation & Support, Multipurpose Development

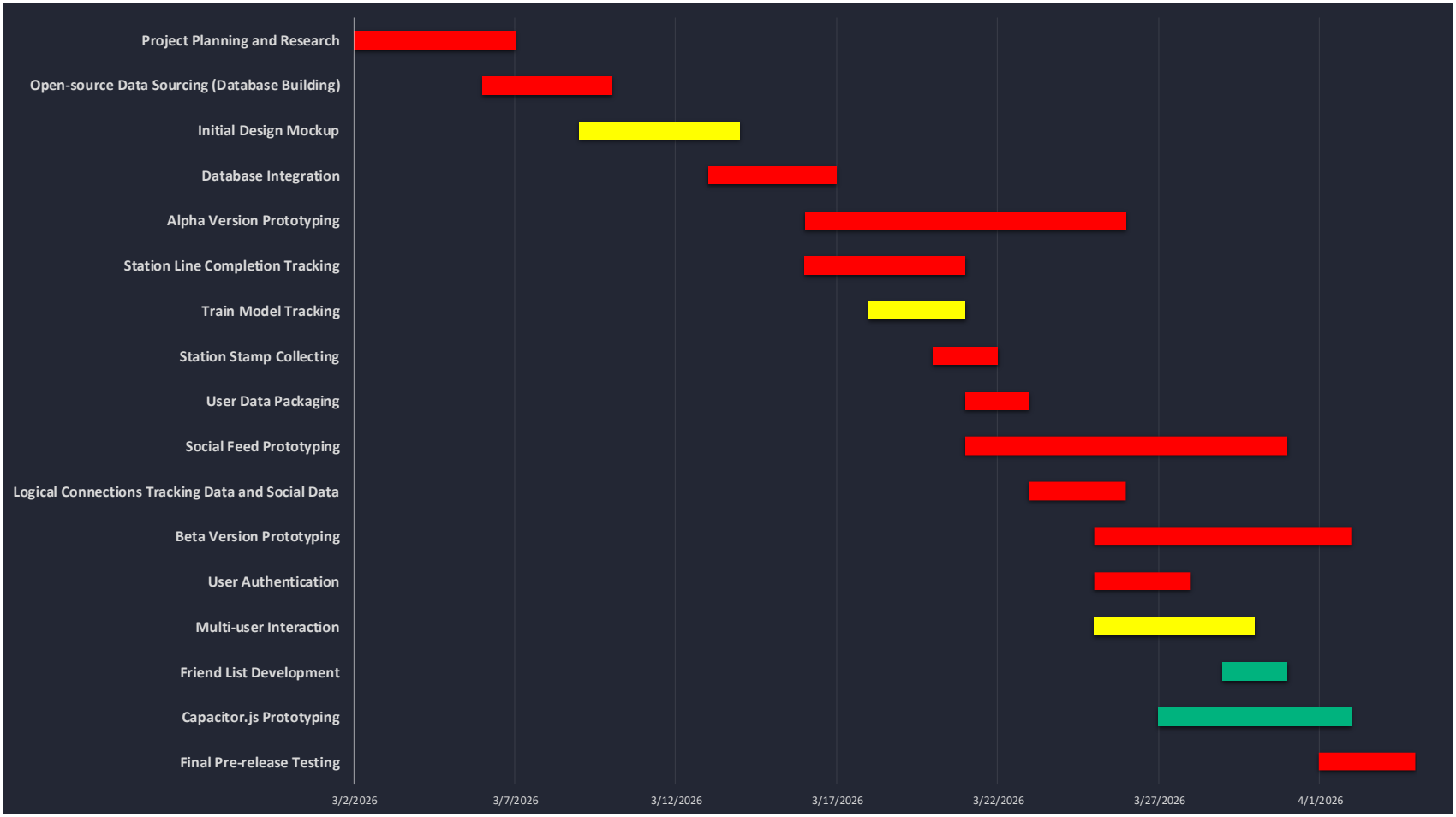
These roles are not strictly defined and will follow a flexible nature in the development environment.

Schedule

Timeline:

Task Name	Start (Date)	End (Date)	Duration (Days)	Priority	Percentage of Completion
Project Planning and Research	02/03/26	06/03/26	5	High	50%
Open-source Data Sourcing (Database Building)	06/03/26	09/03/26	4	High	25%
Initial Design Mockup	09/03/26	13/03/26	5	Medium	25%
Database Integration	13/03/26	16/03/26	4	High	0%
Alpha Version Prototyping	16/03/26	25/03/26	10	High	0%
Station Line Completion Tracking	16/03/26	20/03/26	5	High	0%
Train Model Tracking	18/03/26	20/03/26	3	Medium	0%
Station Stamp Collecting	20/03/26	21/03/26	2	High	0%
User Data Packaging	21/03/26	22/03/26	2	High	0%
Social Feed Prototyping	21/03/26	30/03/26	10	High	0%
Logical Connections Tracking Data and Social Data	23/03/26	25/03/26	3	High	0%
Beta Version Prototyping	25/03/26	01/04/26	8	High	0%
User Authentication	25/03/26	27/03/26	3	High	0%
Multi-user Interaction	25/03/26	29/03/26	5	Medium	0%
Friend List Development	29/03/26	30/03/26	2	Low	0%
Capacitor.js Prototyping	27/03/26	01/04/26	6	Low	0%
Final Pre-release Testing	01/04/26	03/04/26	3	High	0%

Gantt Chart:


Project Planning and Research
Open-source Data Sourcing (Database Building)
Initial Design Mockup
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Alpha Version Prototyping
Station Line Completion Tracking
Train Model Tracking
Station Stamp Collecting
User Data Packaging
Social Feed Prototyping
Logical Connections Tracking Data and Social Data
Beta Version Prototyping
User Authentication
Multi-user Interaction
Friend List Development
Capacitor.js Prototyping
Final Pre-release Testing
3/2/2026
3/7/2026
3/12/2026
3/17/2026
3/22/2026
3/27/2026
4/1/2026

Financial Projections

This summary outlines the projected operational costs and market trends for the 'Train Travel' App based on the 2026 marketplace and the infrastructure required to scale the service.

Infrastructure & API Scaling Projections

As the user base grows, the app may need to transition from Google's free "Spark" plan to a pay-as-you-go model.

- **Google Maps Platform**
  - This app relies on Dynamic Maps for interactive tracking. The Google Maps Platform (2026) provides **10,000 free monthly requests.**
  - Subsequent usage for a base of **50,000 active users** is projected at **$1,500 – $3,000 per month** based on the current rate of **$7.00 per 1,000 requests**.
- **Firebase Backend**
  - Scaling Google Firestore to handle thousands of "Visits," "Sightings," and "Stamps" is estimated to cost **$200 – $500 per month** as document reads and writes increase.

Monetization & Market Trend Sources

The following projections reflect the revenue potential if the app is monetized through advertising or subscriptions to offset operational expenses:

- **Advertising Potential:** With a **CPM (Cost Per Mille [Thousand]) of $5.00 – $10.00**, based on a standard for the Japanese travel sector in 2026, an audience of **100,000 monthly active users** could generate **$36,000 – $60,000 annually**.

- **Subscription Revenue:** Market research indicates for hobbyist apps in Japan ranges from **¥62 to ¥83 per month**. A "Pro" tier at **$2.00/month** with a **5% conversion rate** (5,000 subscribers) would yield **$120,000 annually**.

- **Market Viability:** The Japanese Location-Based Entertainment (LBE) market is valued at **$310 million** and is growing at a **9.6% CAGR**, providing a stable environment for "Station Collection" digital services.

- **Sector Growth:** The broader Japanese railroad market is projected to grow at a **6.7% CAGR through 2030**, ensuring a steady influx of domestic and international users.

Findings and Recommendations

Based on the information within this Feasibility Study Report, these are the conclusions:

Findings and Recommendations

- **Market Viability:** The project successfully identifies a “middle ground” in the Japanese travel market, positioned between navigation apps (Google Maps) and complex RPG games (Ekimemo!). It also addresses the limitations of physical stamp books, which are prone to damage and difficult to search for.

- **Infrastructure Efficiency:** Utilizing Google Firebase and the Google Maps JavaScript API allows for a cost-effective development phase with zero inherent initial costs.

- **Target Audience Alignment:** The app's features such as line completion tracking, digital stamp collecting, and train model logging promote gamified aspects to both train enthusiasts and general tourists.

- **Development Progress:** As of March 4, 2026, the project has completed 50% of its planning and research and 25% of its initial design and data sourcing. A proof-of-concept live site and GitHub repository are already established showing that the project is steadily on track for completion thus far.

- **Technical Scaling & Portability:** While initial development is surrounding a web application, the project should prioritize using the Capacitor JS library as early as possible to port the app to Android and iOS to increase accessibility for travelers on the move in areas where cellular signal may be poor.